Demand

- Demand comes from the behavior of buyers.
- The **quantity demanded** of any good is the amount of the good that buyers are willing and able to purchase.
- **Law of demand**: the claim that the quantity demanded of a good falls when the price of the good rises, other things equal



The Demand Schedule

- **Demand schedule:**
  A table that shows the relationship between the price of a good and the quantity demanded.

The Demand Schedule

- **Demand schedule:**
  A table that shows the relationship between the price of a good and the quantity demanded.
- Example:
  Helen's demand for good X.

Price of lattes	Quantity of X demanded
$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4

Helen's Demand Schedule & Curve

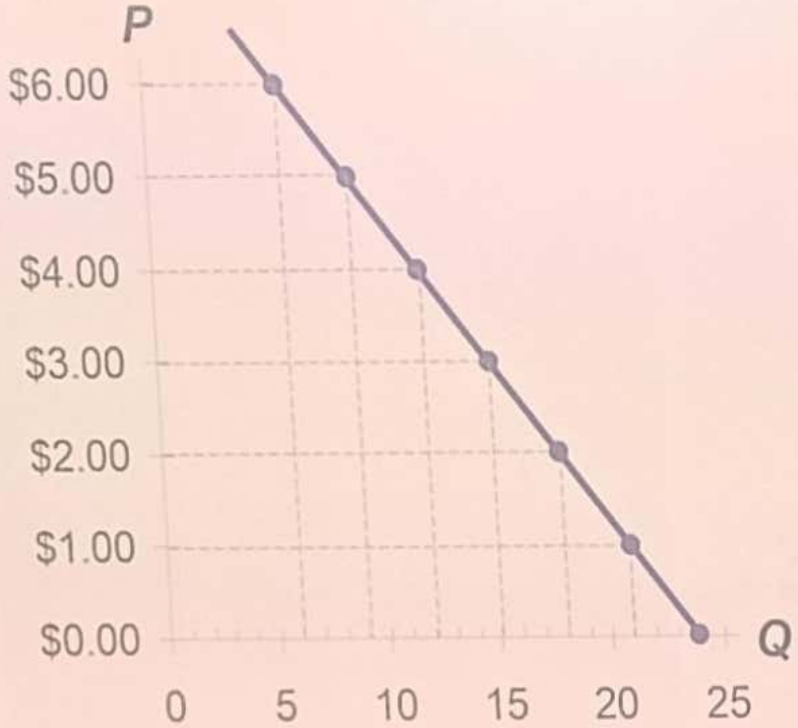
Price of lattes	Quantity of X demanded
$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4

Market Demand versus Individual Demand

- The quantity demanded in the market is the sum of the quantities demanded by all buyers at each price.
- Suppose Helen and Ken are the only two buyers in the Latte market. (Q^d = quantity demanded)

Price	Helen's Q^d	Ken's Q^d	Market Q^d
$0.00	16	8	
1.00	14	7	
2.00	12	6	
3.00	10	5	
4.00	8	4	
5.00	6	3	
6.00	4	2	

The Market Demand Curve for Lattes

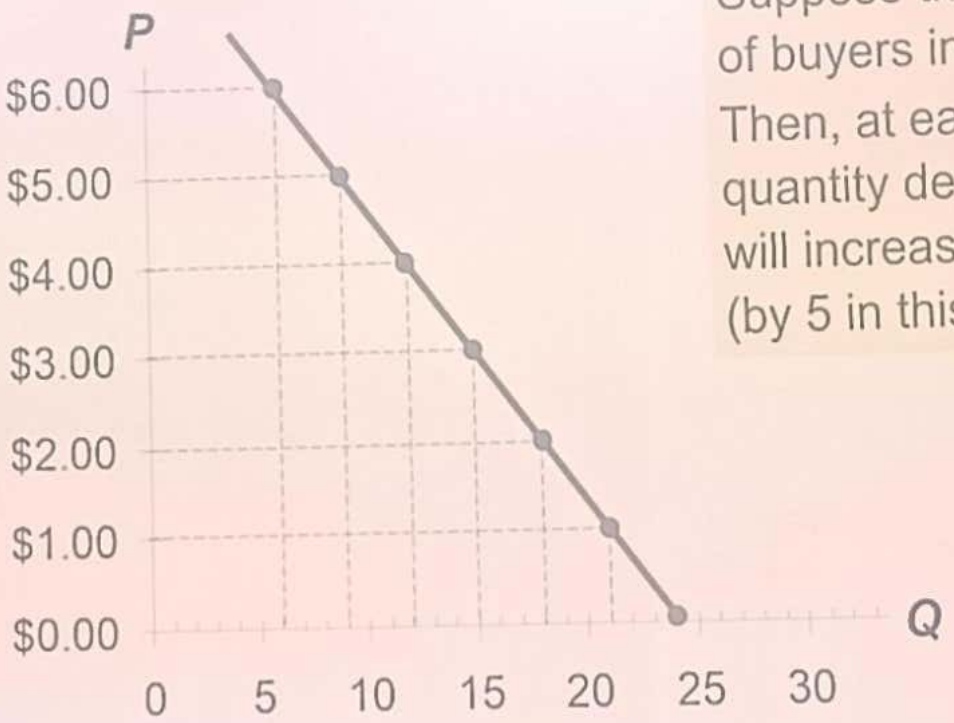


P	Q^d (Market)
$0.00	24
1.00	21
2.00	18
3.00	15
4.00	12
5.00	9
6.00	6

Demand Curve Shifters

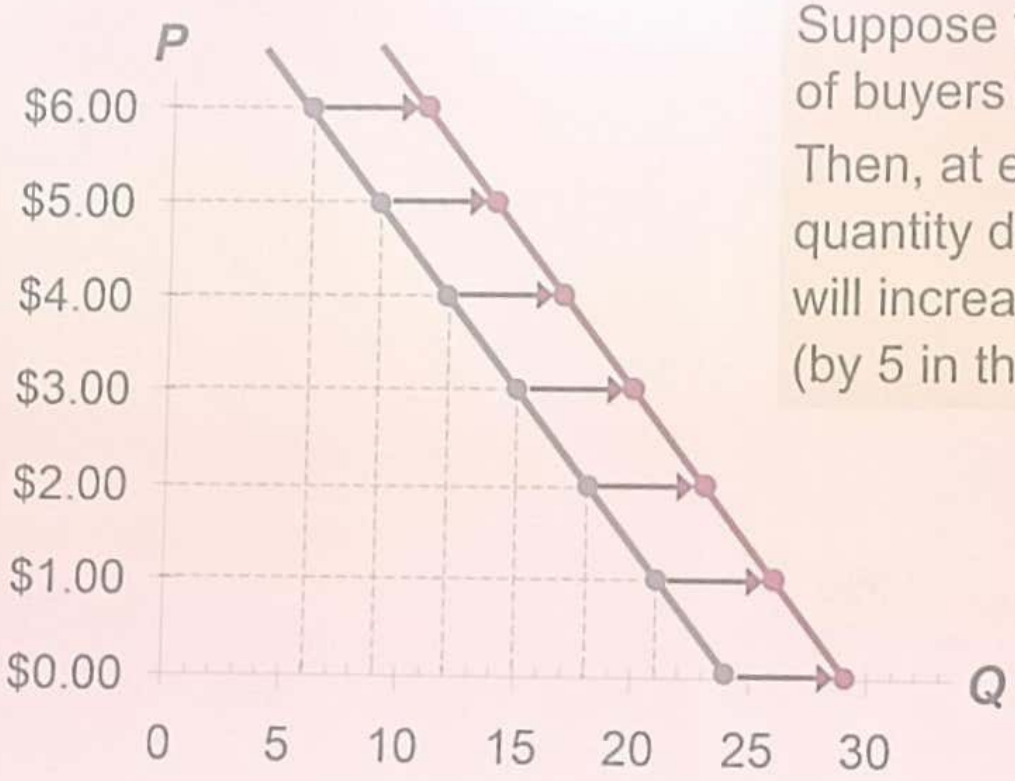
- The demand curve shows how price affects quantity demanded, *other things being equal*.
- These "other things" are non-price determinants of demand (*i.e.*, things that determine buyers' demand for a good, other than the good's price).
- Changes in them shift the **D** curve…

Demand Curve Shifters: # of buyers



Suppose the number of buyers increases.
Then, at each price, quantity demanded will increase (by 5 in this example).

Demand Curve Shifters: # of buyers



Suppose the number of buyers increases.

Then, at each price, quantity demanded will increase (by 5 in this example).

Demand Curve Shifters: income

- Demand for a **normal good** is positively related to income.
  - An increase in income causes increase in quantity demanded at each price, shifting the **D** curve to the right.

(Demand for an **inferior good** is negatively related to income. An increase in income shifts **D** curves for inferior goods to the left.)

Demand Curve Shifters: income

- Demand for a **normal good** is positively related to income.

Demand Curve Shifters: prices of related goods

- Two goods are **substitutes** if
  an increase in the price of one causes
  an increase in demand for the other.
- Example: pizza and hamburgers.
  An increase in the price of pizza
  increases demand for hamburgers,
  shifting hamburger demand curve to the right.
- Other examples: tea and coffee, Coke and Pepsi,
  laptops and desktop computers

Demand Curve Shifters: prices of related goods

- Two goods are **complements** if an increase in the price of one causes a fall in demand for the other.
- Example: computers and software.
  If price of computers rises, people buy fewer computers, and therefore less software.
  Software demand curve shifts left.
- Other examples: college tuition and textbooks, bread and butter etc

Demand Curve Shifters: expectations

- Expectations affect consumers' buying decisions.
- Examples:
  - If people expect their incomes to rise, their demand for meals at expensive restaurants may increase now.
  - If the economy turns bad and people worry about their future job security, demand for new autos may fall now.

Summary: Variables That Affect Demand

Variable	A change in this variable…
Price	…causes a movement along the **D** curve
No. of buyers	…shifts the **D** curve
Income	…shifts the **D** curve
Price of related goods	…shifts the **D** curve
Tastes	…shifts the **D** curve
Expectations	…shifts the **D**

Supply

- Supply comes from the behavior of sellers.
- The **quantity supplied** of any good is the amount that sellers are willing and able to sell.
- **Law of supply**: the claim that the quantity supplied of a good rises when the price of the good rises, other things equal

The Supply Schedule

- **Supply schedule:**
  A table that shows the relationship between the price of a good and the quantity supplied.
  - Example:
    Starbucks' supply of lattes.
  - Notice that Starbucks' supply schedule obeys the Law of Supply.

Price of lattes	Quantity of lattes supplied
$0.00	0
1.00	3
2.00	6
3.00	9
4.00	12
5.00	15
6.00	18

Starbucks' Supply Schedule & Curve



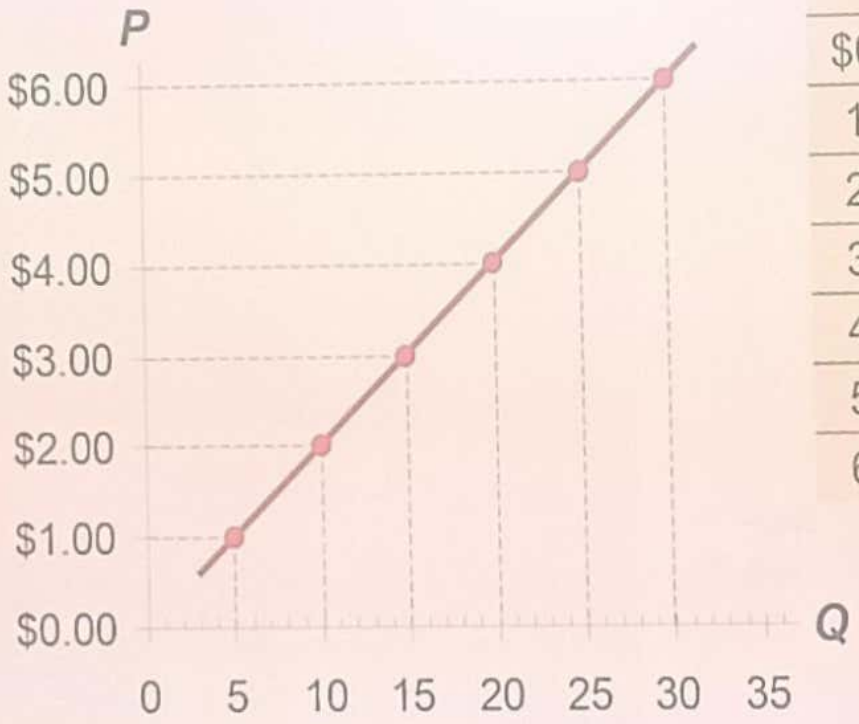
Price of lattes	Quantity of lattes supplied
$0.00	0
1.00	3
2.00	6
3.00	9
4.00	12
5.00	15
6.00	18

Market Supply versus Individual Supply

- The quantity supplied in the market is the sum of the quantities supplied by all sellers at each price.
- Suppose Starbucks and Jitters are the only two sellers in this market. (Q^S = quantity supplied)

Price	Starbucks		Jitters		Market Q^S
$0.00	0	+	0	=	0
1.00	3	+	2		
2.00	6		4		
3.00	9		6		
4.00	12		8		
5.00	15		10		
6.00	18		12		

The Market Supply Curve



P	Q^S (Market)
$0.00	0
1.00	5
2.00	10
3.00	15
4.00	20
5.00	25
6.00	30

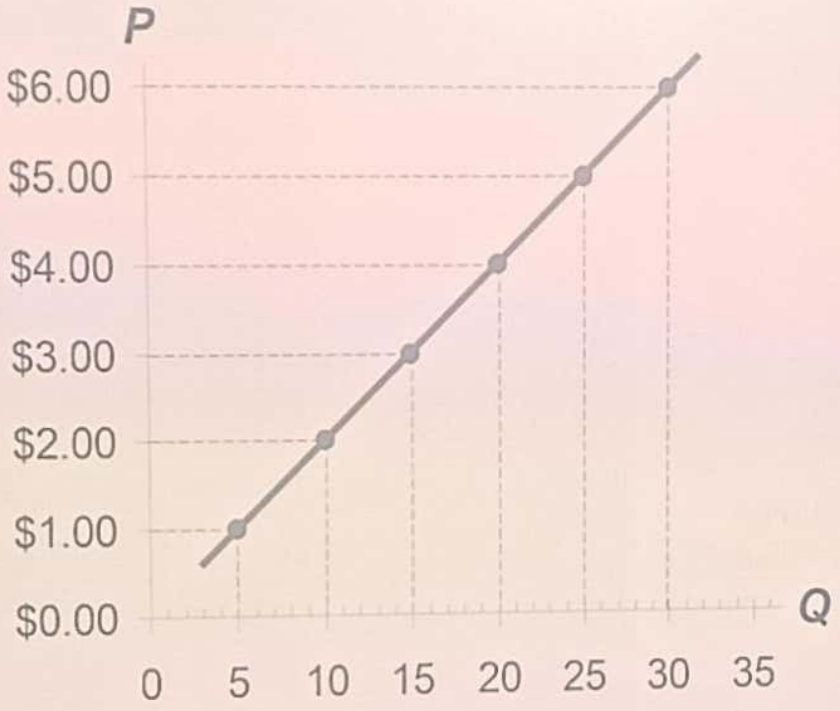
Supply Curve Shifters

- The supply curve shows how price affects quantity supplied, *other things being equal*.
- These "other things" are non-price determinants of supply.
- Changes in them shift the **S** curve…

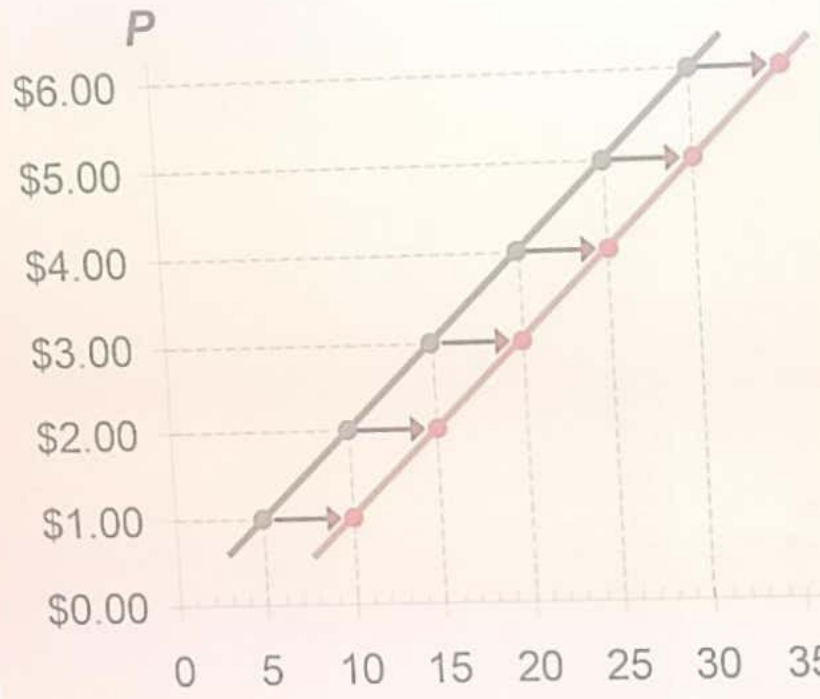
Supply Curve Shifters: input prices

- Examples of input prices:
  wages, prices of raw materials.
- A fall in input prices makes production more profitable at each output price, so firms supply a larger quantity at each price, and the **S** curve shifts to the right.

Supply Curve Shifters: input prices



Supply Curve Shifters: input prices



Suppose the price of milk falls.

At each price, the quantity of Lattes supplied will increase (by 5 in this example).

Supply Curve Shifters: # of sellers

- Technology determines how much inputs are required to produce a unit of output.
- Cost-saving technological improvement has same effect as a fall in input prices, shifting S curve to the right.

Supply Curve Shifters: # of sellers

- An increase in the number of sellers increases the quantity supplied at each price,

Supply Curve Shifters: expectations

- Suppose a firm expects the price of the good it sells to rise in the future.

Supply Curve Shifters: expectations

- Suppose a firm expects the price of the good it sells to rise in the future.
- The firm may reduce supply now, to save some of its inventory to sell later at the higher price.

Supply Curve Shifters: expectations

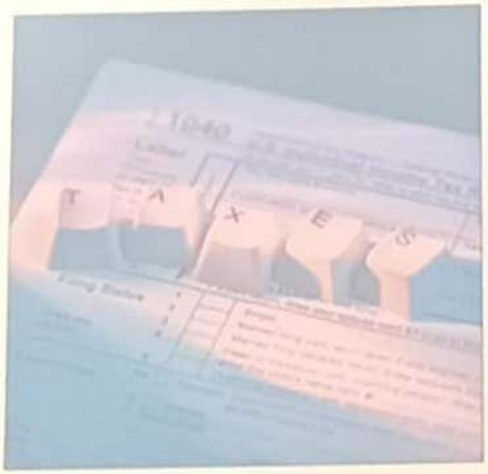
- Suppose a firm expects the price of the good it sells to rise in the future.
- The firm may reduce supply now, to save some of its inventory to sell later at the higher price.
- This would shift the **S** curve leftward.

Summary: Variables That Affect Supply

Variable	A change in this variable…
Price	…causes a movement along the **S** curve
Input prices	…shifts the **S** curve
Technology	…shifts the **S** curve
No. of sellers	…shifts the **S** curve
Expectations	…shifts the **S** curve

ACTIVE LEARNING 2:

Supply curve



Draw a supply curve for tax return preparation software. What happens to it in each of the following scenarios?

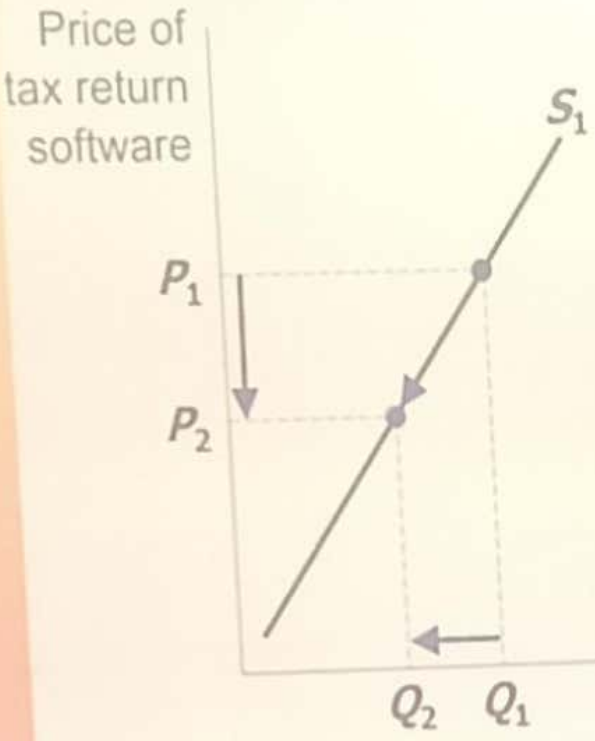
A. Retailers cut the price of the software.

B. A technological advance allows the software to be produced at lower cost.

C. Professional tax return preparers raise the price of the services they provide.

ACTIVE LEARNING 2:

A. fall in price of tax return software

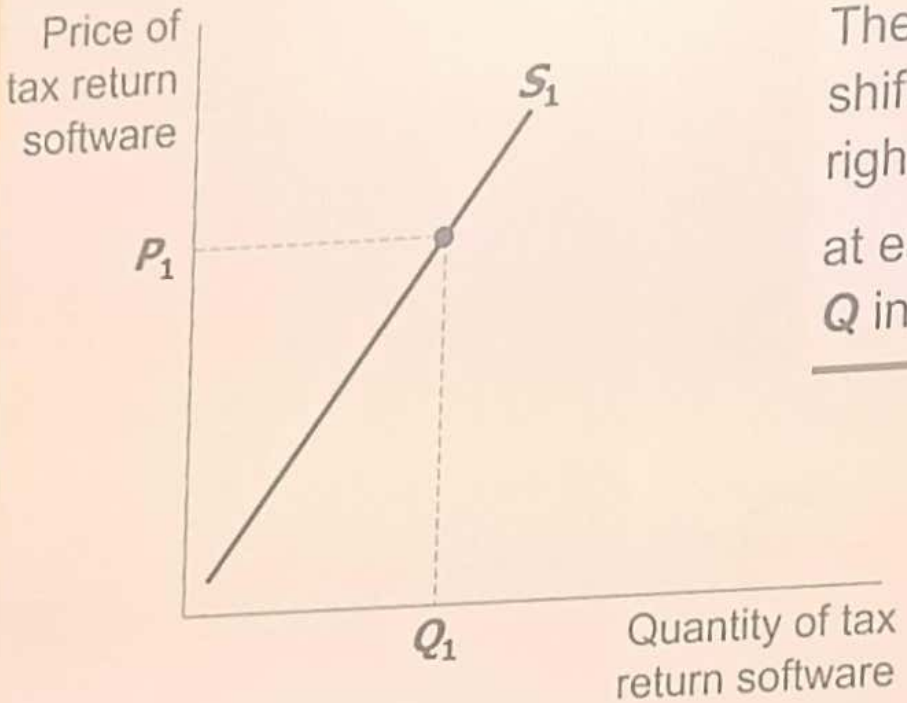


The **S** curve does not shift.

Move down along the curve to a lower *P* and lower *Q*.

ACTIVE LEARNING 2:

B. fall in cost of producing the software

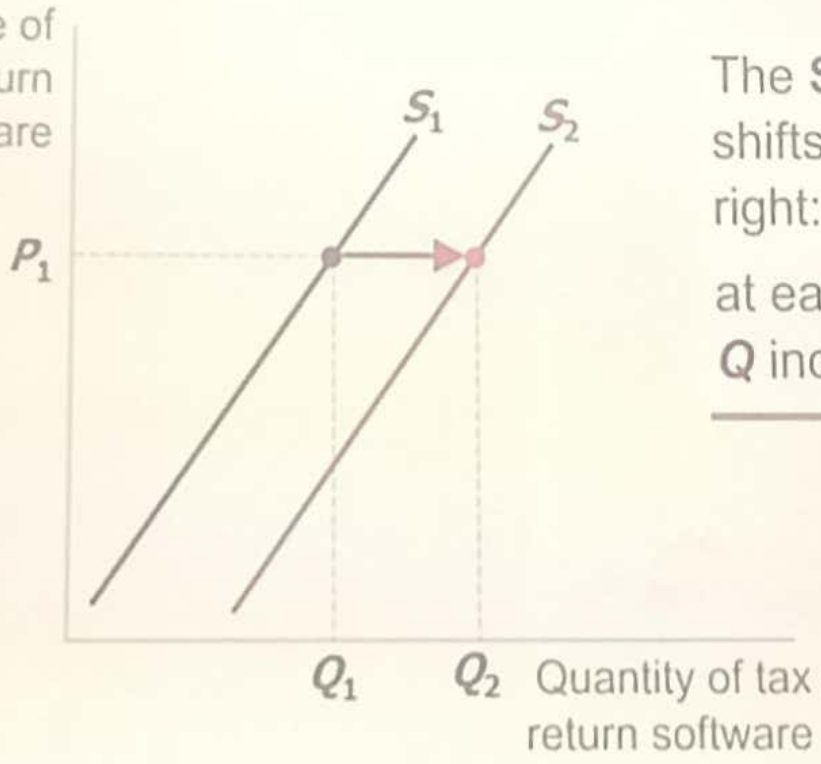


The **S** curve shifts to the right:

at each price, Q increases.

ACTIVE LEARNING 2:

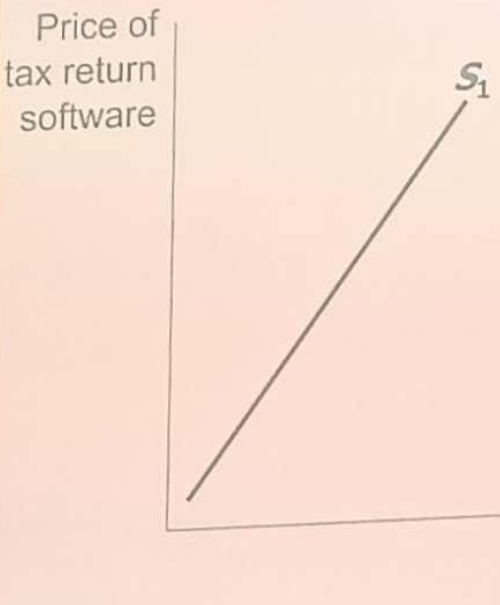
B. fall in cost of producing the software



The **S** curve shifts to the right:

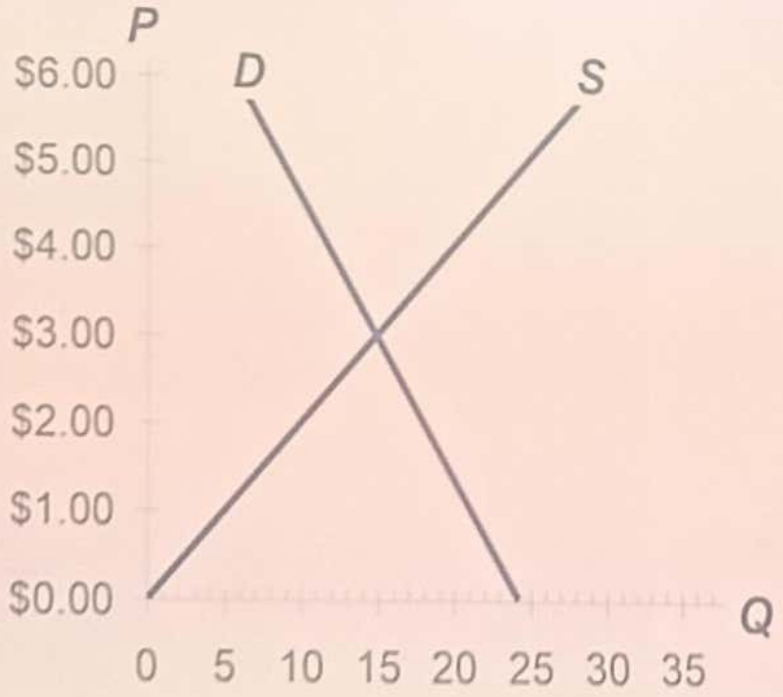
at each price, **Q** increases.

ACTIVE LEARNING 2:
C. professional preparers raise their price



This shifts the demand curve for tax preparation software, not the supply curve.

Supply and Demand Together



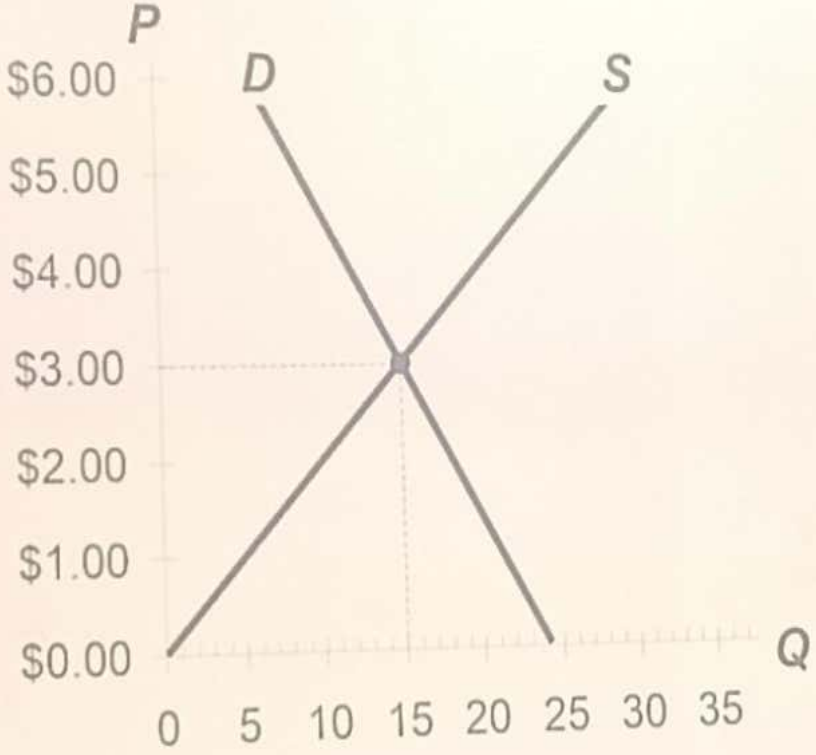
Supply and Demand Together



Equilibrium:
P has reached the level where quantity supplied equals quantity demanded

Equilibrium price:

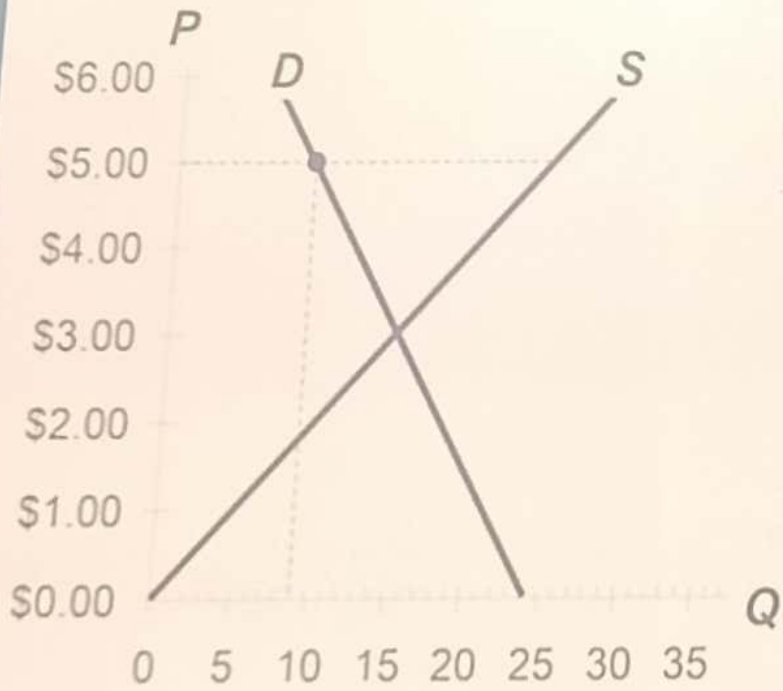
The price that equates quantity supplied with quantity demanded



P	Q^D	Q^S
$0	24	0
1	21	5
2	18	10
3	15	15
4	12	20
5	9	25
6	6	30

Surplus:

when quantity supplied is greater than quantity demanded



Example:
If P = \$5,

then
Q^D = 9 lattes